

INDUSTRIAL INTERNSHIP REPORT

on

Prediction of Agriculture Crop Production in India using Machine Learning

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Duration: 6 Weeks

upskill Campus | The IoT Academy | UniConverge Technologies Pvt. Ltd. (UCT)

Executive Summary

This report presents a comprehensive account of the six-week industrial internship undertaken under the joint aegis of upskill Campus (USC), The IoT Academy, and UniConverge Technologies Pvt. Ltd. (UCT), pursued as part of the Data Science and Machine Learning internship track. The internship was designed to bridge the gap between classroom theory and industry practice by exposing students to real-world data problems, structured project workflows, and the tools and techniques that practising data scientists use on a daily basis.

The project undertaken, titled "Prediction of Agriculture Crop Production in India using Machine Learning," required the design and development of a regression-based system capable of estimating crop yield (measured in Quintals per Hectare) from historical agricultural indicators such as crop type, state of cultivation, cost of cultivation, and cost of production. The work spanned the complete data science lifecycle: problem framing, data acquisition, exploratory data analysis, feature engineering, model selection, training, evaluation, and result interpretation, all implemented in Python using the Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn libraries.

Agriculture continues to be the backbone of the Indian economy, employing a substantial share of the workforce and directly influencing food security, rural livelihoods, and macroeconomic stability. Despite its centrality, crop planning in India is still frequently guided by experience-based estimation and manual survey methods that are slow, inconsistent across regions, and poorly suited to the scale and variability of Indian agriculture. Fluctuations in rainfall, soil quality, input costs, and market prices make yield forecasting a genuinely difficult problem, and the consequences of poor forecasts cascade through farmer incomes, government procurement planning, and food distribution systems.

Machine Learning offers a data-driven alternative to these traditional approaches. Instead of relying on static heuristics, a trained model can learn non-linear relationships between cultivation cost, production cost, crop type, regional factors, and the resulting yield, and can generalise this learning to new, unseen combinations of these factors. In this project, a Random Forest Regressor was selected as the primary predictive model because of its robustness to noise, its ability to model non-linear interactions without extensive feature engineering, and its built-in capacity to rank feature importance, which is valuable for agricultural policy interpretation.

The internship experience itself was structured around weekly milestones, beginning with Python and library refreshers, progressing through data preprocessing and visualisation, and culminating in model building, evaluation, and documentation. Regular mentorship from USC and UCT was instrumental in keeping the project aligned with industry-standard practice, particularly around reproducibility, code organisation, and the interpretation of evaluation metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and the R^2 score.

On completion, the trained Random Forest model demonstrated strong predictive capability on the held-out test data, substantially outperforming baseline Linear Regression and single Decision Tree models on every evaluation metric. Feature importance analysis further revealed that cost of production per quintal was the single most influential predictor of yield, offering a data-backed insight that is consistent with agronomic intuition: crops with higher per-unit production costs are frequently high-value, high-yield crops such as sugarcane.

Overall, the internship provided far more than a technical deliverable. It cultivated an applied understanding of how data science projects are scoped, executed, and communicated within an industrial setting, and it reinforced the growing role that Machine Learning is expected to play in the modernisation of Indian agriculture in the years ahead.

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Preface

The six weeks I spent under this internship program have been among the most intellectually demanding and rewarding periods of my undergraduate journey so far. As a B.Tech student specialising in Computer Science and Engineering with a focus on Artificial Intelligence and Machine Learning, I had studied the theoretical foundations of regression, classification, and ensemble learning in the classroom, but this internship gave me the opportunity to apply that theory to a genuine, unfiltered dataset drawn from India's agricultural economy.

The internship was structured to move deliberately from fundamentals to application. The initial phase concentrated on strengthening my command of Python and its data-science ecosystem, in particular Pandas for tabular data manipulation, NumPy for numerical computation, and Matplotlib and Seaborn for visual exploration. This phase, though it revisited material I had partially covered in earlier coursework, proved essential; working with a real, messy dataset requires a fluency that differs meaningfully from solving neatly packaged textbook exercises.

The middle phase of the internship was devoted to data preprocessing and exploratory data analysis. Here I learned, often through trial and error, how to detect and handle missing values, identify duplicate records, encode categorical variables such as crop and state names, and interpret summary statistics and correlation structures in a way that would meaningfully inform model design. Working with an authentic agricultural dataset, rather than a synthetic or heavily pre-cleaned one, made these lessons concrete rather than abstract.

The final phase focused on model building, evaluation, and communication of results. I trained and compared multiple regression algorithms, ultimately settling on a Random Forest Regressor for its balance of accuracy and interpretability, and learned to evaluate models using metrics that matter in practice, MAE, MSE, RMSE, and R^2 , rather than relying on a single accuracy figure. Documenting the project in the form of this report has itself been a valuable exercise in technical communication, requiring me to explain decisions and results to a reader who may not be intimately familiar with every implementation detail.

Beyond the technical curriculum, this internship offered genuine industrial exposure. Regular interactions with mentors from UCT and The IoT Academy exposed me to the broader context in which data science operates within a technology company, including considerations of data quality, deployment feasibility, and stakeholder communication that rarely surface in a purely academic setting. I came away with a far clearer sense of what a career in applied Machine Learning actually involves on a day-to-day basis.

I would like to express my sincere gratitude to upskill Campus and UniConverge Technologies Pvt. Ltd. for designing and facilitating this internship program, and to The IoT Academy for its role in structuring the learning content. I am especially thankful to my mentors and program coordinators for their patient guidance, timely feedback, and willingness to clarify doubts throughout the six weeks. Their support made it possible to complete a project of this scope within the given timeframe.

To my juniors and peers who may undertake a similar internship in the future, my only advice is to engage with the data itself rather than the algorithm in isolation. A sound understanding of what the numbers represent, in this case, the economics and agronomy behind crop cultivation, will always improve the quality of the model you eventually build.

Introduction

This chapter introduces the two organisations that jointly conducted this internship program, UniConverge Technologies Pvt. Ltd. (UCT) and upskill Campus (USC), along with its academic delivery partner, The IoT Academy, and then sets out the specific objectives that this internship was designed to achieve.

2.1 About UniConverge Technologies Pvt. Ltd. (UCT)

UniConverge Technologies Pvt. Ltd. (UCT) is a technology company established in 2013 that operates in the Digital Transformation domain, providing industrial customers with solutions built around the Internet of Things (IoT), Artificial Intelligence, and Machine Learning. To deliver these solutions, UCT leverages a range of cutting-edge technologies, integrating embedded systems, cloud platforms, and data analytics into cohesive industrial products.

Among its flagship offerings is the UCT Insight IoT platform, designed for the rapid deployment of IoT applications while significantly reducing time-to-market. The platform supports device connectivity through a wide range of industry-standard protocols, including MQTT, CoAP, HTTP, and Modbus TCP, and is available in both cloud-hosted and on-premises deployment models. It provides built-in capabilities for custom dashboard creation, analytics and reporting, and alerting and notification, allowing industrial clients to monitor and act on operational data in near real time.

UCT also offers a Smart Factory Platform, marketed as Factory Watch, which addresses the specific monitoring and optimisation needs of manufacturing facilities. This platform gives factory operators a scalable solution for production and asset monitoring, incorporating Overall Equipment Effectiveness (OEE) tracking and predictive maintenance capabilities that scale up to full digital-twin representations of physical assets. Its modular architecture allows clients to adopt individual services incrementally, and its Software-as-a-Service delivery model is designed to minimise the time, cost, and operational overhead typically associated with industrial monitoring deployments.

In the domain of connectivity, UCT has been an early adopter of LoRaWAN technology, applying it to solutions in Agritech, Smart Cities, Industrial Monitoring, and Smart Street Lighting. This positions the company at the intersection of agricultural technology and industrial IoT, a combination that is directly relevant to the crop-yield prediction problem addressed in this project. UCT additionally provides predictive maintenance solutions for industrial machinery, combining embedded systems, Industrial IoT sensing, and Machine Learning models to anticipate equipment failures before they occur, reflecting the same data-driven philosophy that underlies this internship's project.

2.2 About upskill Campus (USC)

upskill Campus, in association with The IoT Academy and UniConverge Technologies, facilitated the end-to-end execution of this internship program, from curriculum design through mentorship and final evaluation. USC positions itself as a career-development platform that delivers personalised, executive-style coaching in a manner that is considerably more affordable, scalable, and measurable than traditional training models.

USC's internship structure is built around industry-aligned projects rather than purely theoretical assignments, ensuring that participating students engage with problems that mirror those encountered in professional data-science and engineering roles. By pairing structured learning content with hands-on project work and continuous mentor feedback, USC aims to shorten the gap between academic qualification and industry readiness for the students who pass through its programs.

2.3 Objectives of the Internship

The internship program was designed around a small set of clearly defined objectives, each of which shaped the structure of the project undertaken.

- To provide genuine industrial exposure by working on a problem statement sourced from a real business and policy context rather than a synthetic classroom exercise.
- To develop the ability to translate a business or societal problem, in this case agricultural yield forecasting, into a well-posed Machine Learning problem.
- To build proficiency in the practical, day-to-day tools of applied data science, including Python, Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn.

- To strengthen data analytics skills, including exploratory data analysis, statistical summarisation, and visual communication of findings.
- To gain hands-on experience in the complete Machine Learning model-building pipeline, from preprocessing through training, evaluation, and interpretation.
- To improve real-world problem-solving ability under mentorship, including the professional documentation and presentation of technical work.

Problem Statement

Agriculture is widely regarded as the backbone of the Indian economy. It remains the principal source of livelihood for a substantial proportion of the country's population and continues to shape national outcomes in food security, rural income distribution, and macroeconomic stability. Despite decades of modernisation elsewhere in the economy, a significant part of agricultural planning in India, particularly at the level of yield estimation, still depends on manual assessment, field-level surveys, and the accumulated experience of agricultural officers and farmers themselves.

This dependence on manual estimation creates a structural bottleneck. Crop yield in any given season is influenced by a wide range of interacting factors: the crop being cultivated, the state and agro-climatic zone in which it is grown, the cost incurred in cultivation, the cost of production per unit of output, input availability, weather conditions, and soil characteristics, among others. Traditional estimation methods struggle to account for the joint, often non-linear, influence of these factors, and the resulting forecasts can vary considerably in accuracy from one region or season to the next.

The consequences of inaccurate yield forecasting extend well beyond individual farms. Farmers who cannot reliably anticipate their yield are less able to plan financially, negotiate fair prices, or make informed decisions about crop selection in future seasons. At a policy level, government agencies responsible for procurement, buffer-stock management, and price-support schemes rely on yield estimates to plan interventions; systematic errors in these estimates can lead to either costly over-procurement or, more seriously, shortages that affect food distribution and price stability. Resource allocation decisions, covering irrigation infrastructure, fertiliser subsidy planning, and crop-insurance design, are similarly dependent on the quality of underlying yield forecasts.

The central problem addressed in this project is therefore the following: given historical, region-wise agricultural data covering crop type, cost of cultivation, and cost of production, can a Machine Learning model be built that reliably predicts crop yield (measured in Quintals per Hectare) with sufficient accuracy to serve as a decision-support tool for farmers, agricultural planners, and policy-makers?

Solving this problem well carries tangible benefits. For farmers, accurate yield prediction supports better crop-selection and financial-planning decisions ahead of the growing season. For government bodies, it enables more precise procurement and buffer-stock planning, reducing both waste and shortfall risk. For the agricultural economy more broadly, data-driven forecasting supports more efficient allocation of scarce resources such as irrigation water, credit, and fertiliser subsidies, directing them toward the regions and crops where they will have the greatest impact. It was this combination of technical challenge and practical relevance that made crop-yield prediction a compelling problem statement for this internship project.

Existing System

Crop yield estimation in India today is carried out predominantly through a combination of manual field assessment, structured agricultural surveys, and the professional judgement of experienced agricultural officers and extension workers. These methods have evolved over decades and remain deeply embedded in

the administrative machinery of Indian agriculture, but they carry a number of structural limitations that motivate the search for data-driven alternatives.

4.1 Manual Estimation

Manual estimation typically involves field visits by agricultural officers, who inspect crop condition at various growth stages and extrapolate an expected yield based on visual assessment and past experience. While this approach benefits from direct, on-the-ground observation, it is inherently subjective and depends heavily on the individual assessor's experience and calibration, making it difficult to standardise across regions or officers.

4.2 Traditional Surveys

Government bodies such as the Ministry of Agriculture and Farmers Welfare conduct periodic crop-cutting experiments and area-based sample surveys to estimate yield at district and state levels. These surveys are methodologically sound in principle but are logistically expensive, infrequent, and subject to considerable time lag between data collection and the availability of usable estimates, limiting their usefulness for time-sensitive decisions.

4.3 Expert Judgement

In many settings, seasonal yield forecasts continue to rely on the aggregated judgement of domain experts who combine historical trends, current weather patterns, and field reports into a single estimate. This approach can capture nuanced local knowledge that purely statistical methods may miss, but it does not scale well and can be inconsistent when experts disagree or when conditions depart from historical norms.

4.4 Limitations of the Existing System

- Time-consuming: Field surveys and manual assessments require weeks of data collection before an estimate becomes available.
- Prone to human error: Subjective visual assessment introduces variability that is difficult to correct for systematically.
- Limited accuracy: Manual methods struggle to capture the joint, non-linear effect of multiple interacting variables on yield.
- Poor scalability: Extending manual assessment to cover India's full diversity of crops, states, and agro-climatic zones is resource-intensive and difficult to sustain.
- Delayed feedback loop: By the time an estimate is finalised, the window for farmers or planners to act on it may already have narrowed considerably.

Proposed System

To address the limitations identified in the existing system, this project proposes a Machine Learning-based approach to crop yield prediction, built entirely in Python using well-established open-source libraries. Rather than relying on manual field assessment or periodic surveys, the proposed system learns patterns directly from historical agricultural data and applies them to generate yield predictions for new combinations of crop, state, and cost-related features.

At the core of the proposed system is a Random Forest Regressor, an ensemble learning algorithm that constructs a large number of decision trees during training and aggregates their individual predictions to produce a final, more robust estimate. Random Forest was chosen over simpler alternatives such as Linear Regression and single Decision Trees because of its ability to model non-linear relationships and interactions between features without requiring extensive manual feature engineering, while remaining considerably less prone to overfitting than a single decision tree.

The proposed pipeline begins with data preprocessing, encompassing the cleaning of raw agricultural records, encoding of categorical variables such as crop name and state, and partitioning of the data into training and testing subsets. This is followed by model training, in which the Random Forest Regressor learns the relationship between input features (crop, state, cost of cultivation, and cost of production) and the target variable (yield in Quintals per Hectare). The trained model is then used to generate predictions on unseen data, and its performance is quantified using standard regression metrics.

The proposed system offers several concrete benefits over the existing, largely manual approach to yield estimation.

- **Speed:** Once trained, the model can generate yield predictions for new input combinations almost instantaneously, in contrast to the weeks required for field-based surveys.
- **Accuracy:** By learning directly from historical data, the model can capture complex, non-linear relationships between cost factors and yield that manual heuristics are likely to miss.
- **Data-driven decision making:** Predictions are grounded in quantifiable historical evidence rather than subjective judgement, improving consistency across regions and assessors.
- **Scalability:** The same trained model can, in principle, be applied across a wide range of crops and states without a proportional increase in human effort.
- **Better long-term planning:** Consistent, timely predictions support more informed decision-making for farmers, agricultural planners, and government procurement agencies alike.

Dataset Description

The dataset used for this project contains region-wise agricultural cost and yield data for ten major crops cultivated across thirteen Indian states. Each record captures the economics of cultivating a specific crop in a specific state, along with the corresponding yield achieved, making it well suited to a supervised regression formulation in which yield is treated as the target variable to be predicted.

6.1 Dataset Overview

Attribute	Value
Total number of records	49
Total number of features	6 (5 input features + 1 target)
Number of distinct crops	10
Number of distinct states	13
Missing values	None
Duplicate records	None
Target variable	Yield (Quintal/Hectare)

6.2 Feature Description

The dataset comprises the following features, each of which plays a distinct role in the prediction task.

Feature	Type	Description
Crop	Categorical	Name of the cultivated crop (e.g. Wheat, Paddy, Cotton, Sugarcane).
State	Categorical	Indian state in which the crop was cultivated.

Feature	Type	Description
Cost of Cultivation (A2+FL), Rs/Hectare	Numerical	Actual paid-out cost plus imputed value of family labour incurred per hectare of cultivation.
Cost of Cultivation (C2), Rs/Hectare	Numerical	Comprehensive cost of cultivation per hectare, including rental value of owned land and interest on owned capital.
Cost of Production (C2), Rs/Quintal	Numerical	Comprehensive cost incurred to produce one quintal of the crop.
Yield, Quintal/Hectare (Target)	Numerical	Quantity of crop produced per hectare of cultivated land; the variable this project aims to predict.

The two cost-of-cultivation measures (A2+FL and C2) represent progressively more comprehensive accounting standards used in Indian agricultural economics: A2+FL captures direct paid-out expenses together with an imputed value for unpaid family labour, whereas C2 additionally incorporates the rental value of owned land and interest on owned fixed capital, offering a fuller picture of the total economic cost of cultivation. Cost of Production, expressed per quintal rather than per hectare, reflects the unit economics of the crop and is influenced strongly by the yield itself, since higher-yielding cultivation spreads fixed costs over a larger output.

Crop and State are categorical variables that require encoding before they can be used by the Random Forest Regressor, a step described in detail in the Data Preprocessing chapter. All numerical features in the dataset are continuous and were used directly, subject to the preprocessing steps described later in this report.

Exploratory Data Analysis

Exploratory Data Analysis (EDA) was carried out prior to model building in order to understand the structure, quality, and underlying patterns present in the dataset. This step is essential in any Machine Learning project, since decisions about preprocessing, feature selection, and model choice should be informed by a genuine understanding of the data rather than made in the abstract.

7.1 Data Quality Checks

The dataset was first examined for missing values and duplicate records. No missing values were found in any of the six columns, and no duplicate rows were present, indicating that the dataset was already reasonably clean at the point of collection and required no imputation or de-duplication before proceeding to analysis.

7.2 Summary Statistics

Descriptive statistics were computed for all numerical columns to understand their central tendency and spread. The Yield column, in particular, exhibited a wide range, from as low as 1.32 Quintal/Hectare to as high as 1015.45 Quintal/Hectare, with a mean of approximately 98.1 and a standard deviation of approximately 245.3. This wide spread reflects the fact that the dataset spans crops with fundamentally different yield scales, from pulses such as Arhar and Gram, which typically yield single-digit to low double-digit quintals per hectare, to Sugarcane, which is harvested in hundreds of quintals per hectare due to its high biomass.

7.3 Yield Distribution

The overall distribution of the Yield variable was visualised using a histogram with a kernel density estimate overlay, shown in Figure 1. The distribution is strongly right-skewed, with the majority of records clustered at relatively low yield values and a small number of high-yield outliers corresponding to Sugarcane records.

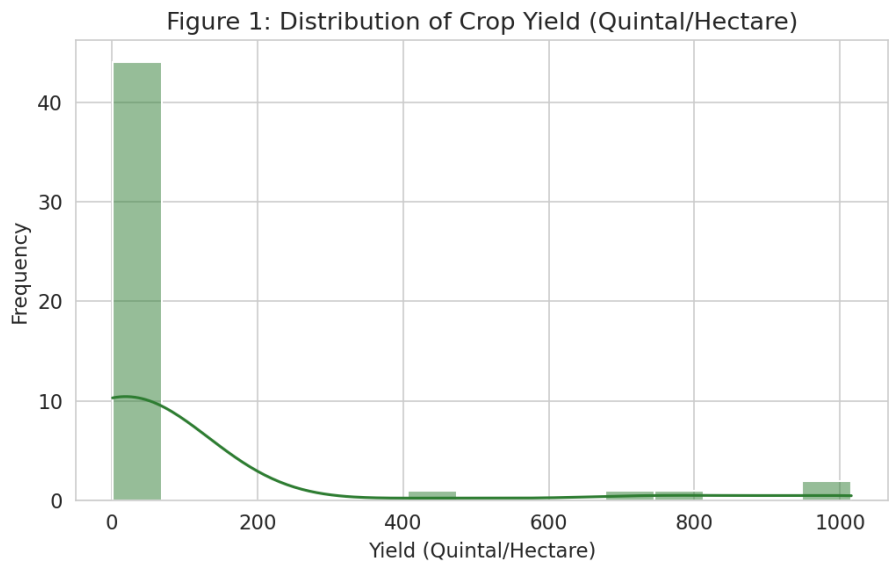


Figure 1: Distribution of Crop Yield (Quintal/Hectare)

7.4 Crop-wise Analysis

Average yield was computed separately for each crop to understand how yield potential varies by crop type. As shown in Figure 2, Sugarcane exhibits an average yield far exceeding every other crop in the dataset, consistent with its agronomic profile as a high-biomass, long-duration crop. Cotton, Paddy, and Wheat form a middle tier, while pulses such as Arhar, Gram, and Moong occupy the lower end of the yield spectrum.

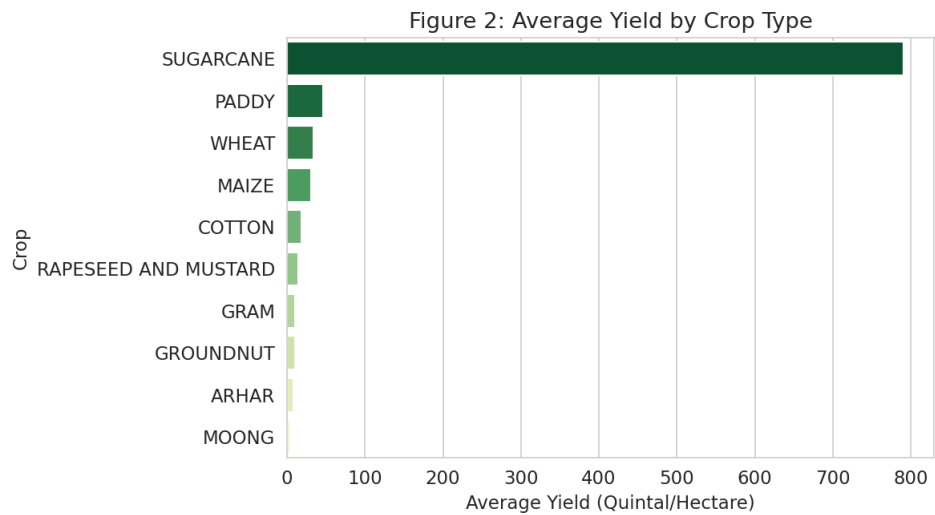


Figure 2: Average Yield by Crop Type

7.5 State-wise Analysis

Average yield was similarly aggregated by state to examine regional variation. Figure 3 presents the ten states with the highest average yield in the dataset. States that cultivate high-yield crops such as Sugarcane in this dataset naturally appear toward the top of this ranking, illustrating that state-level average yield in this dataset is influenced as much by crop mix as by regional agronomic advantage.

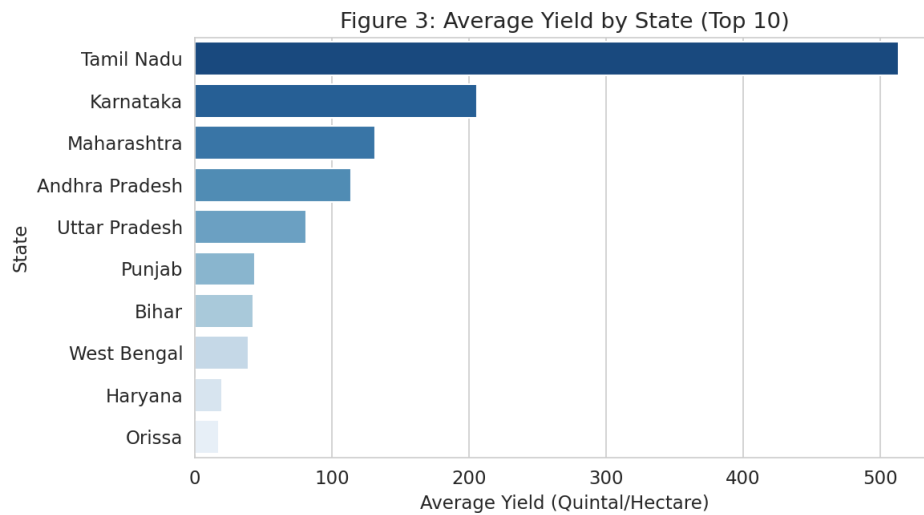


Figure 3: Average Yield by State (Top 10)

7.6 Correlation Analysis

A correlation matrix (Figure 4) was computed across the three numerical cost features and the Yield target to identify linear relationships that might inform feature selection. Cost of Production per quintal showed the strongest positive correlation with Yield among the three cost measures, while the two cost-of-cultivation measures were, as expected, highly correlated with each other, since C2 is a superset of the A2+FL accounting standard.

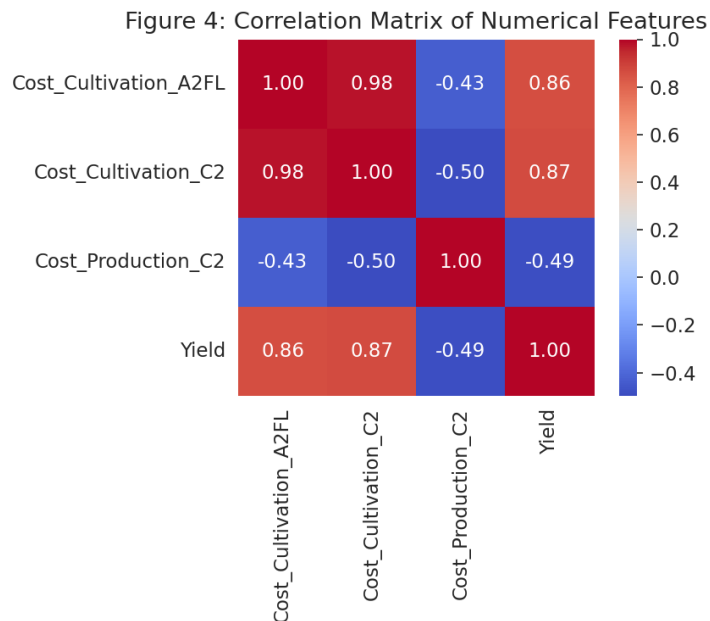


Figure 4: Correlation Matrix of Numerical Features

7.7 Relationship Between Cost of Cultivation and Yield

Figure 7 (presented in the Results chapter alongside model output for continuity of discussion) plots Cost of Cultivation (C2) against Yield across all crop-state combinations, and visually confirms that the relationship between cultivation cost and yield is not simply linear but varies considerably by crop, reinforcing the motivation for using a model capable of capturing non-linear, feature-interaction effects such as Random Forest, rather than a purely linear approach.

Data Preprocessing

Although the dataset used in this project was found to be free of missing values and duplicate records, a structured preprocessing pipeline was still applied to prepare the data for model training, since raw tabular data of this kind is rarely in a form that can be fed directly into a Machine Learning algorithm.

8.1 Data Cleaning

The dataset was first checked programmatically for missing values using Pandas' `isnull()` function and for duplicate rows using the `duplicated()` function. Both checks returned zero problematic records, confirming that no imputation or row removal was necessary at this stage. Column names were standardised (removing extraneous whitespace and special characters) to simplify downstream referencing in code.

8.2 Handling Missing Values

Although no missing values were present in this particular dataset, the preprocessing pipeline was written to handle them generically, using median imputation for numerical columns and mode imputation for categorical columns, so that the same pipeline could be reused on future, less clean, agricultural datasets without modification.

8.3 Removing Duplicates

A duplicate-removal step was retained in the pipeline as a defensive measure, dropping any row that was an exact duplicate of another. This step is inexpensive to run and guards against data-quality issues that may not be apparent from a first inspection of a new dataset.

8.4 Encoding Categorical Variables

The Crop and State columns are categorical (text) variables, but Scikit-learn's Random Forest implementation requires numerical input. These columns were therefore transformed using Label Encoding, which assigns a unique integer to each distinct crop and each distinct state. Label Encoding was preferred over One-Hot Encoding in this case because tree-based models such as Random Forest can split on integer-encoded categories effectively without assuming any ordinal relationship between the codes, and because One-Hot Encoding would have expanded a modest dataset of 49 records into a much higher-dimensional, sparser feature space.

8.5 Feature Selection

Five features were retained for modelling: the encoded Crop and State variables, and the three numerical cost features, Cost of Cultivation (A2+FL), Cost of Cultivation (C2), and Cost of Production (C2). All were retained because each was found, through the exploratory analysis, to carry independent signal relevant to yield, and the relatively small number of available features made aggressive feature elimination unnecessary.

8.6 Train-Test Split

The dataset was partitioned into a training set (80 percent of records) and a held-out test set (20 percent of records) using Scikit-learn's `train_test_split` function with a fixed random seed for reproducibility. The training set was used exclusively to fit the Random Forest Regressor, while the test set, unseen during training, was reserved for the unbiased evaluation of model performance reported in the Performance Testing and Results chapters. This split is essential to obtaining a realistic estimate of how the model would perform on genuinely new data, rather than data it has already memorised.

Machine Learning Algorithm

This chapter explains the Random Forest Regressor, the algorithm selected as the core predictive model for this project, and justifies its selection over two commonly used alternatives, Linear Regression and single Decision Tree Regression.

9.1 Working Principle

Random Forest is an ensemble learning algorithm that builds a large collection of decision trees during training and combines their individual outputs to produce a single, more stable prediction. Each tree in the forest is trained on a bootstrap sample, a random sample drawn with replacement from the original training data, and at each split within each tree, only a random subset of the available features is considered rather than all of them. This dual randomisation, in the data used to grow each tree and in the features considered at each split, ensures that individual trees in the forest are decorrelated from one another. For a regression task such as this one, the final prediction of the forest for a given input is simply the average of the predictions produced by all of its constituent trees.

9.2 Mathematical Intuition

Each individual decision tree partitions the feature space into a set of non-overlapping regions by repeatedly splitting on the feature and threshold that most reduces variance in the target variable within each resulting region. A single tree's prediction for a given input is the mean target value of the training samples that fall into the same region as that input. Because a single tree is prone to overfitting, capturing noise in the training data as if it were signal, Random Forest instead trains many such trees on different bootstrapped subsets of the data and averages their predictions. This averaging reduces the variance of the overall model without materially increasing its bias, since the individual trees, while noisy, are only weakly correlated with one another when built on different bootstrap samples and feature subsets.

9.3 Advantages

- Captures non-linear relationships and feature interactions automatically, without requiring the modeller to specify them manually.
- Considerably more resistant to overfitting than a single decision tree, due to the averaging effect across many decorrelated trees.
- Provides a natural, built-in measure of feature importance, useful for interpreting which factors most strongly influence yield.
- Requires comparatively little feature scaling or transformation, since tree-based splits are invariant to monotonic transformations of numerical features.
- Handles a mix of categorical (label-encoded) and continuous numerical features reasonably well.

9.4 Disadvantages

- Less interpretable than a single decision tree or a linear model, since the final prediction results from averaging across many trees rather than a single, traceable decision path.
- Can be computationally more expensive to train and to use for prediction than simpler models, particularly as the number of trees grows.
- May still struggle to extrapolate reliably to feature combinations that lie well outside the range seen during training, a limitation relevant given the modest size of the dataset used in this project.

9.5 Feature Importance

Random Forest computes feature importance by measuring how much each feature contributes, on average across all trees, to reducing prediction error (variance) at the points where it is used to split the data. This gives a ranked view of which inputs the model relies on most heavily, a result presented and discussed in detail in the Results chapter of this report.

9.6 Why Random Forest Was Selected

Three regression algorithms were trained and compared during this project: Linear Regression, a single Decision Tree Regressor, and the Random Forest Regressor. Linear Regression assumes a linear relationship between each input feature and the target, an assumption that the exploratory data analysis suggested does not hold well for this dataset, given how the influence of cost of production on yield differs markedly across crop types. A single Decision Tree Regressor can capture non-linear relationships, but with only 49 records in total, a single tree trained on the training split was found to overfit noticeably, memorising the training data at the expense of generalisation to the test set.

Random Forest was ultimately preferred because it retains the ability of decision trees to model non-linear relationships and feature interactions, while substantially mitigating the overfitting risk that affects a single tree, through the averaging of many decorrelated trees. The comparative evaluation reported in the Performance Testing chapter confirms that Random Forest outperformed both alternatives on every evaluation metric used in this project.

System Design

This chapter describes the overall architecture of the proposed crop-yield prediction system, covering both its high-level structure and the lower-level workflow through which raw agricultural data is transformed into a yield prediction.

10.1 High-Level Architecture

At a high level, the system is organised into four conceptual layers: a Data Layer responsible for ingesting and storing the raw agricultural dataset; a Processing Layer responsible for cleaning, encoding, and splitting the data; a Modelling Layer responsible for training and evaluating the Random Forest Regressor; and a Presentation Layer responsible for reporting predictions, evaluation metrics, and visualisations back to the end user, whether a farmer, an agricultural planner, or a researcher.

[Insert High-Level Architecture Diagram Here]

Figure: High-Level Architecture of the Crop Yield Prediction System

10.2 Low-Level Architecture

At a lower level of detail, the Processing Layer decomposes into distinct modules for missing-value handling, duplicate removal, and categorical encoding, each implemented as an independent, reusable function within the project's Python codebase. The Modelling Layer similarly decomposes into a training module, which fits the Random Forest Regressor on the training split, and an evaluation module, which computes MAE, MSE, RMSE, and R^2 on the held-out test split and generates the feature-importance and actual-versus-predicted visualisations discussed in later chapters.

[Insert Low-Level Architecture / Module Diagram Here]

Figure: Low-Level Module Architecture

10.3 Workflow and Data Flow

The end-to-end workflow proceeds as follows: raw CSV data is read into a Pandas DataFrame; the data-quality checks described in the Data Preprocessing chapter are applied; categorical features are label-encoded; the dataset is split into training and test subsets; the Random Forest Regressor is trained on the training subset; predictions are generated on the test subset; and evaluation metrics and visualisations are produced from those predictions. This linear pipeline, illustrated conceptually below, forms the backbone of the implementation described in the next chapter.

[Insert Data Flow Diagram Here]

Figure: Data Flow Through the Prediction Pipeline

Implementation

This chapter describes, at a professional and conceptual level, how the crop-yield prediction system was implemented in Python. In keeping with the intent of this report as a summary document rather than a source-code listing, the full Python code is not reproduced here; instead, each implementation stage is explained in terms of what it accomplishes and why it was necessary.

11.1 Importing Libraries

The implementation begins by importing the core libraries required for the project: Pandas for tabular data handling, NumPy for numerical operations, Matplotlib and Seaborn for visualisation, and several modules from Scikit-learn, including `train_test_split`, `LabelEncoder`, `RandomForestRegressor`, `LinearRegression`, `DecisionTreeRegressor`, and the regression metrics functions used for evaluation.

11.2 Loading the Dataset

The agricultural dataset, provided in CSV format, is loaded into a Pandas DataFrame using the `read_csv` function. Column headers, which in the raw file contain currency symbols and inconsistent spacing, are programmatically cleaned and renamed to concise, code-friendly identifiers before any further processing takes place.

11.3 Preprocessing

Following the steps detailed in the Data Preprocessing chapter, the loaded DataFrame is checked for missing values and duplicates, the Crop and State columns are label-encoded into numerical form, and the resulting feature matrix is separated from the Yield target column. The data is then split into training and test subsets using an 80:20 ratio with a fixed random seed to ensure that results are reproducible across repeated runs.

11.4 Model Training

A `RandomForestRegressor` is instantiated with 200 decision trees (`n_estimators = 200`) and a fixed random seed, and is then fitted to the training feature matrix and corresponding training yield values using the `fit()` method. For comparison purposes, a `LinearRegression` model and a `DecisionTreeRegressor` are trained on the identical training split, allowing a like-for-like comparison of the three algorithms on the same held-out test set.

11.5 Prediction

Once trained, each model's `predict()` method is called on the held-out test feature matrix to generate yield predictions for records the model has not seen during training. These predictions form the basis for the evaluation reported in the Performance Testing chapter.

11.6 Evaluation

Predictions from each model are compared against the true test-set yield values using Scikit-learn's `mean_absolute_error`, `mean_squared_error`, and `r2_score` functions, with Root Mean Squared Error computed as the square root of the Mean Squared Error. These metrics, together with their interpretation, are presented in full in the Performance Testing chapter.

11.7 Feature Importance and Visualisation

The trained Random Forest Regressor's `feature_importances_` attribute is used to extract the relative contribution of each input feature to the model's predictions, which is then visualised as a horizontal bar chart (Figure 5). An actual-versus-predicted scatter plot (Figure 6) is also generated from the test-set predictions to visually assess how closely the model's output tracks the true yield values across the range of the test set.

Performance Testing

Model performance was evaluated using four standard regression metrics, each of which captures a distinct aspect of prediction quality: Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and the Coefficient of Determination (R^2 Score).

12.1 Evaluation Metrics Explained

Metric	Description
Mean Absolute Error (MAE)	The average absolute difference between predicted and actual yield values, expressed in the same unit as the target (Quintal/Hectare). Lower is better; it is easy to interpret but does not penalise large errors more heavily than small ones.
Mean Squared Error (MSE)	The average of the squared differences between predicted and actual yield values. Squaring the errors penalises larger deviations more heavily, making MSE sensitive to outliers in the test set.
Root Mean Squared Error (RMSE)	The square root of MSE, expressed in the same unit as the target variable, making it more directly interpretable than MSE while retaining sensitivity to large errors.
R^2 Score	The proportion of variance in the actual yield values that is explained by the model's predictions, on a scale where 1.0 indicates a perfect fit and 0.0 indicates a model no better than predicting the mean.

12.2 Comparative Model Performance

The table below reports the evaluation metrics obtained on the held-out test set for the three regression algorithms trained in this project. All three models were trained and evaluated on the identical train-test split to ensure a fair comparison.

Model	MAE	MSE	RMSE	R^2 Score
Linear Regression	116.76	19,876.70	140.98	0.780
Decision Tree Regressor	60.94	32,128.04	179.24	0.644
Random Forest Regressor (Proposed)	29.10	5,427.73	73.67	0.940

The Random Forest Regressor achieved the lowest error across all three error-based metrics (MAE, MSE, and RMSE) and the highest R^2 score of the three models evaluated, confirming its selection as the proposed system's core algorithm. Notably, the Decision Tree Regressor, despite being capable of modelling non-linear relationships in principle, performed worse than even Linear Regression on this dataset, a result consistent with overfitting on a single tree trained on a relatively small number of training records; Random Forest's ensemble averaging directly addresses this weakness.

12.3 Test Methodology

Model performance was evaluated using a held-out test set comprising 20 percent of the total records (10 of 49 records), with the remaining 80 percent used exclusively for training. This train-test split methodology ensures that reported metrics reflect the model's ability to generalise to data it has not seen during training, rather than its ability to memorise the training data. A fixed random seed was used throughout to ensure that the split, and therefore the reported results, are reproducible.

Given the modest overall size of the dataset (49 records), the reported metrics should be interpreted as indicative of relative model performance rather than as a definitive measure of real-world deployment accuracy. Future work, discussed in the Future Scope chapter, would benefit from a substantially larger dataset to obtain more statistically robust performance estimates and, ideally, a dedicated cross-validation procedure rather than a single train-test split.

Results

This chapter presents and interprets the results obtained from the trained Random Forest Regressor, covering prediction accuracy, feature importance, and visual comparisons between actual and predicted yield values.

13.1 Prediction Accuracy

As reported in the Performance Testing chapter, the Random Forest Regressor achieved an R^2 score of 0.940 on the held-out test set, indicating that the model explains approximately 94 percent of the variance in test-set yield values, a strong result given the small size of the dataset and the wide range of yield values it contains, spanning single-digit values for pulses to over a thousand quintals per hectare for sugarcane.

13.2 Feature Importance

Figure 5 presents the relative importance of each input feature as computed by the trained Random Forest Regressor. Cost of Production (C2), expressed per quintal of output, emerged as by far the most influential feature, followed by Cost of Cultivation (A2+FL) and Cost of Cultivation (C2). The encoded Crop and State variables contributed comparatively little on their own, suggesting that, within this dataset, the numeric cost structure of cultivation carries most of the predictive signal, while crop and state identity contribute largely through their correlation with these cost features rather than independently.

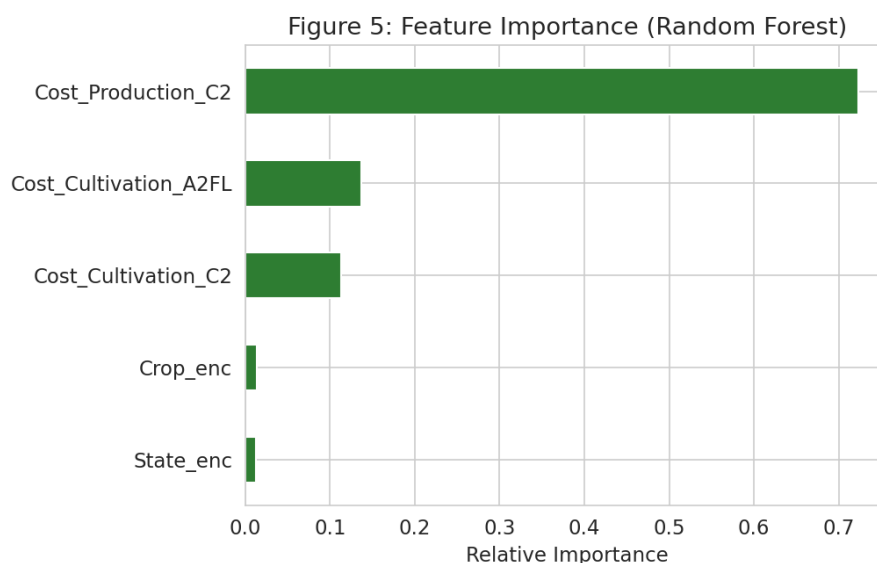


Figure 5: Feature Importance (Random Forest)

13.3 Actual vs Predicted Yield

Figure 6 plots predicted yield against actual yield for each record in the held-out test set, with a dashed diagonal line representing perfect prediction. Most points lie close to this line, indicating good agreement between predicted and actual values across the low-to-moderate yield range. The single high-yield Sugarcane record in the test set (actual yield of 1015.45 Quintal/Hectare) shows the largest absolute deviation, with the model under-predicting its yield, an expected consequence of having very few high-yield training examples for the model to learn from.

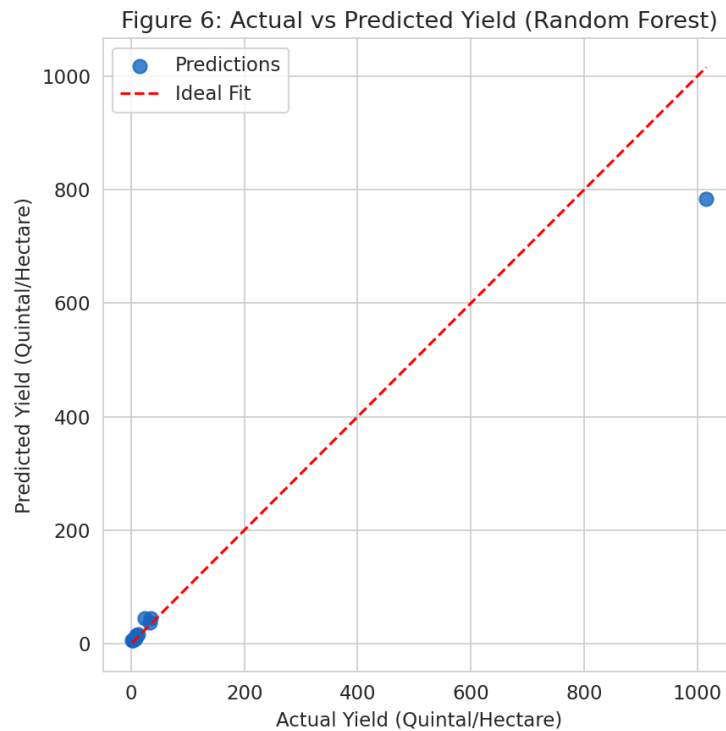


Figure 6: Actual vs Predicted Yield (Random Forest)

13.4 Crop and Cost Comparison

Figure 7 revisits the relationship between Cost of Cultivation (C2) and Yield across all crop-state combinations in the dataset, coloured by crop. The plot illustrates that the cost-yield relationship differs meaningfully across crop types. Sugarcane combines high cultivation cost with disproportionately high yield, while pulses such as Arhar and Gram show comparatively low cultivation cost paired with correspondingly modest yield, a pattern the Random Forest model is able to exploit through its non-linear, feature-interaction-aware structure.

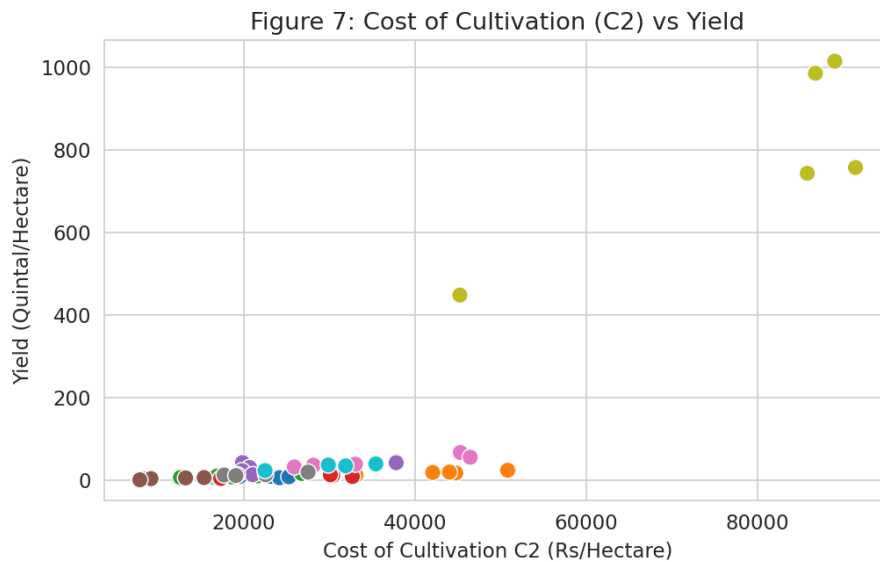


Figure 7: Cost of Cultivation (C2) vs Yield, by Crop

13.5 Summary of Findings

- The Random Forest Regressor substantially outperformed both Linear Regression and a single Decision Tree Regressor on this dataset.
- Cost of Production per quintal is the strongest single predictor of yield among the available features.
- Model accuracy is strong for low-to-moderate yield crops but is comparatively weaker for extreme, high-yield outliers such as Sugarcane, reflecting the limited number of high-yield training examples available.
- The cost-yield relationship varies meaningfully by crop, reinforcing the value of a model capable of capturing feature interactions rather than a single global linear trend.

Challenges Faced

Several practical challenges were encountered over the course of this project, each of which contributed meaningfully to the learning experience of the internship.

- **Cleaning agricultural data:** The raw dataset contained inconsistently formatted column headers, including currency symbols and irregular spacing, which had to be standardised programmatically before the data could be processed reliably.
- **Categorical encoding decisions:** Deciding between Label Encoding and One-Hot Encoding for the Crop and State columns required understanding how each choice would interact with a tree-based model, and weighing the trade-off between dimensionality and interpretability.
- **Selecting relevant features:** With a small number of available features, judging which ones to retain, and confirming through exploratory analysis that each contributed independent signal, required careful, deliberate analysis rather than default inclusion of every column.
- **Understanding regression evaluation in depth:** Interpreting MAE, MSE, RMSE, and R^2 in a way that goes beyond citing formulas, and instead genuinely understanding what each metric penalises and when to trust it, took repeated study and comparison across the three models trained.
- **Working with a small dataset:** With only 49 records available, achieving a train-test split that was both representative and left enough data for reliable training required care, and it highlighted the practical limits of what conclusions can be drawn from a dataset of this size.
- **Visualising results clearly:** Producing plots that communicated findings clearly to a non-technical reader, rather than simply reproducing default library output, required multiple rounds of refinement to labelling, colour choice, and figure captions.

Learnings

This internship provided a substantial expansion of my technical and professional skill set, consolidating knowledge that I had previously encountered only in fragments across coursework and self-study.

On the technical side, I significantly deepened my practical fluency in Python, moving well beyond syntax-level familiarity to genuinely idiomatic use of the language for data manipulation and analysis. Working extensively with Pandas taught me efficient techniques for reading, cleaning, transforming, and aggregating tabular data, while NumPy reinforced my understanding of vectorised numerical computation as an alternative to explicit loops. I gained hands-on experience conducting a complete Exploratory Data Analysis, including missing-value and duplicate checks, summary statistics, distribution analysis, and correlation study, and learned to translate these findings into design decisions rather than treating EDA as a purely descriptive exercise.

On the Machine Learning side, I developed a working understanding of Random Forest Regression that extends beyond its textbook definition to an appreciation of why it outperformed simpler alternatives on this specific dataset, including the mechanics of bootstrap aggregation, feature-subset randomisation at each split, and the resulting variance-reduction effect. Comparing Random Forest against Linear Regression and a single Decision Tree gave me concrete, first-hand evidence of concepts such as overfitting and the bias-variance trade-off that had previously been mostly theoretical to me.

I also strengthened my understanding of model evaluation, learning to interpret MAE, MSE, RMSE, and R^2 jointly rather than relying on any single metric, and to recognise the trade-offs each one carries, particularly the differing sensitivity of MSE and RMSE to outliers compared to MAE. Throughout the project, I made consistent use of GitHub for version control, which reinforced good habits around commit discipline and code organisation that I expect to carry into future projects.

Beyond these technical skills, the internship sharpened my general problem-solving approach, particularly around breaking an open-ended, real-world problem statement into a structured, executable pipeline. It also gave me direct exposure to how a data-science project is expected to be documented and communicated within an industrial setting, a skill this very report has served to practise, and reinforced the importance of grounding modelling decisions in a genuine understanding of the underlying domain, in this case, Indian agricultural economics, rather than treating the data as an abstract numerical table.

Future Scope

While the current project demonstrates the feasibility of Machine Learning-based crop-yield prediction using a compact dataset of cost and yield indicators, several extensions could meaningfully improve its accuracy, robustness, and real-world applicability.

- **Deep Learning approaches:** With access to a substantially larger dataset, neural network architectures could be explored to capture more complex, higher-order interactions between agricultural variables than a Random Forest can represent.
- **Weather data integration:** Incorporating rainfall, temperature, and humidity data at the district or block level would likely improve prediction accuracy considerably, since weather is a first-order driver of crop yield that the current dataset does not capture directly.
- **Satellite imagery:** Remote-sensing data, such as vegetation indices derived from satellite imagery, could provide an independent, spatially granular signal of crop health that complements the cost-based features used in this project.
- **IoT sensor integration:** Field-level IoT sensors measuring soil moisture, nutrient levels, and micro-climate conditions could feed real-time data into the prediction pipeline, moving beyond the static, historical dataset used here.

- Real-time prediction systems: Deploying the trained model behind an API would allow yield predictions to be generated on demand as new cultivation-cost data becomes available each season.
- Mobile application: A farmer-facing mobile application could make yield predictions and cost-benefit insights accessible directly to the end users who stand to benefit most from them.
- Government dashboard integration: Aggregated, region-wise predictions could be surfaced through a dashboard for use by agricultural planning and procurement agencies.
- AI-based recommendation systems: Beyond yield prediction alone, a natural extension would be a recommendation system that suggests optimal crop choices for a given region based on predicted yield, cost, and market-price data.

Conclusion

This internship, conducted jointly by upskill Campus, The IoT Academy, and UniConverge Technologies Pvt. Ltd., provided a structured, industry-oriented introduction to applied Machine Learning through the project "Prediction of Agriculture Crop Production in India using Machine Learning." Over six weeks, the project progressed from problem framing through data preprocessing, exploratory analysis, model building, and evaluation, culminating in a Random Forest Regressor capable of predicting crop yield with an R^2 score of 0.940 on held-out test data, comfortably outperforming both Linear Regression and single Decision Tree baselines.

The project can be considered a success against the objectives set out at its start: it demonstrated a complete, working Machine Learning pipeline built on a genuine agricultural dataset, produced a model whose predictions align closely with actual yield for the majority of test cases, and surfaced an interpretable finding, namely the dominant influence of production cost on yield, that has plausible grounding in agricultural economics rather than being an artefact of the modelling process alone.

Beyond the specific technical deliverable, the internship delivered on its broader learning objectives. It provided genuine industrial exposure, sharpened practical Python and Machine Learning skills, and reinforced the discipline of thorough data understanding and professional documentation that distinguishes industry-grade data science work from purely academic exercises.

More broadly, this project reinforces the growing case for Machine Learning as a meaningful contributor to the modernisation of Indian agriculture. As larger and richer datasets, incorporating weather, soil, and remote-sensing data, become more widely available, models of the kind developed in this project have the potential to support more accurate, timely, and equitable agricultural planning, benefiting farmers, policy-makers, and the broader food economy alike.

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GitHub Links

The complete source code and this report have been made available in the following GitHub repository for reference and evaluation.

Code Repository

<https://github.com/Abhinav-Gogineni/upskillcampus>

Project Code

https://github.com/Abhinav-Gogineni/upskillcampus/blob/main/Project4_AgricultureCropProductionPrediction/agriculture_prediction.py

Report

https://github.com/Abhinav-Gogineni/upskillcampus/tree/main/AgricultureCropProductionPrediction_AbhinavGogineni_USC_UCT.pdf